

Captura de Diagramas Esquemáticos:

Creación de librerías

Basado en materiales elaborados para Electratraining (www.electratraining.org) por:
Miguel Albero Gil (malbero@ai2.upv.es)

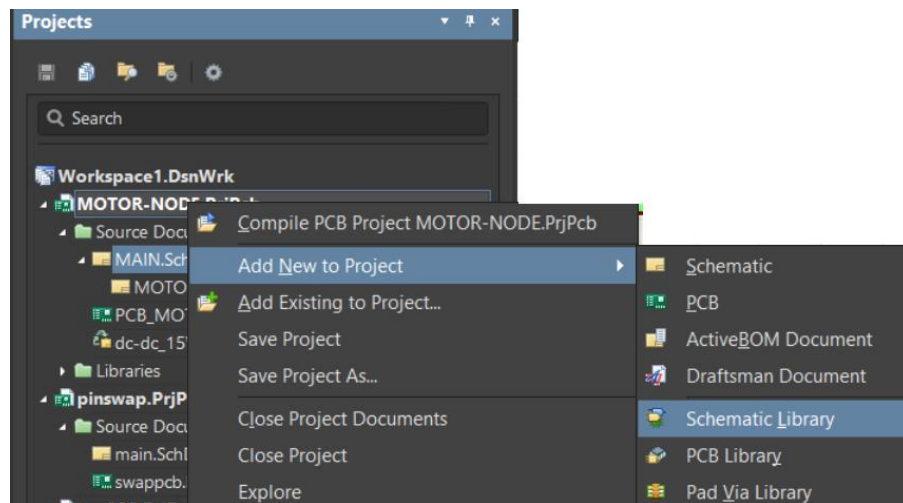
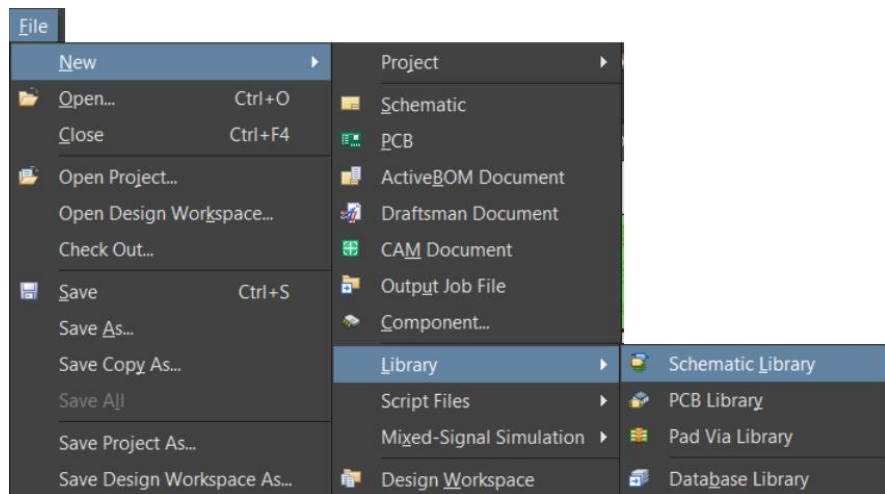


Agenda

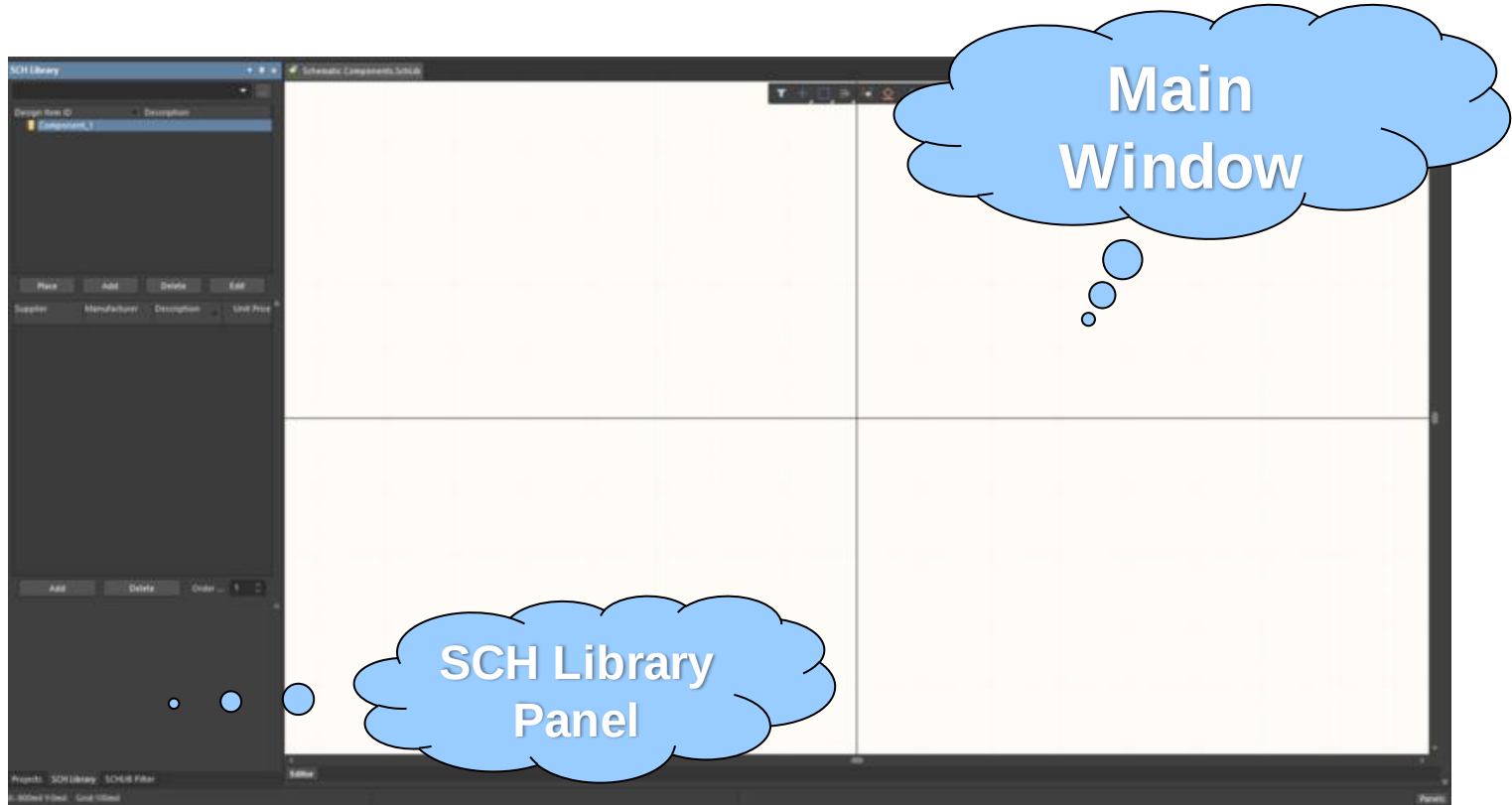
- **SCH libraries**
- PCB libraries
- Via libraries

Creating a New Library

File ➔ New ➔ Library ➔ Schematic Library, or from Project Panel



New Library



New Component

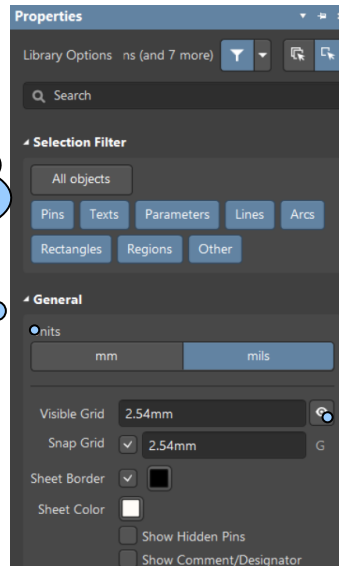
- **Tools ➔ New Component**

- A new library always contains one empty component, simply rename Component_1 (**Tools ➔ Rename Component**)

- Important to set options: **Tools ➔ Document Options**

Imperial or
Metric Units

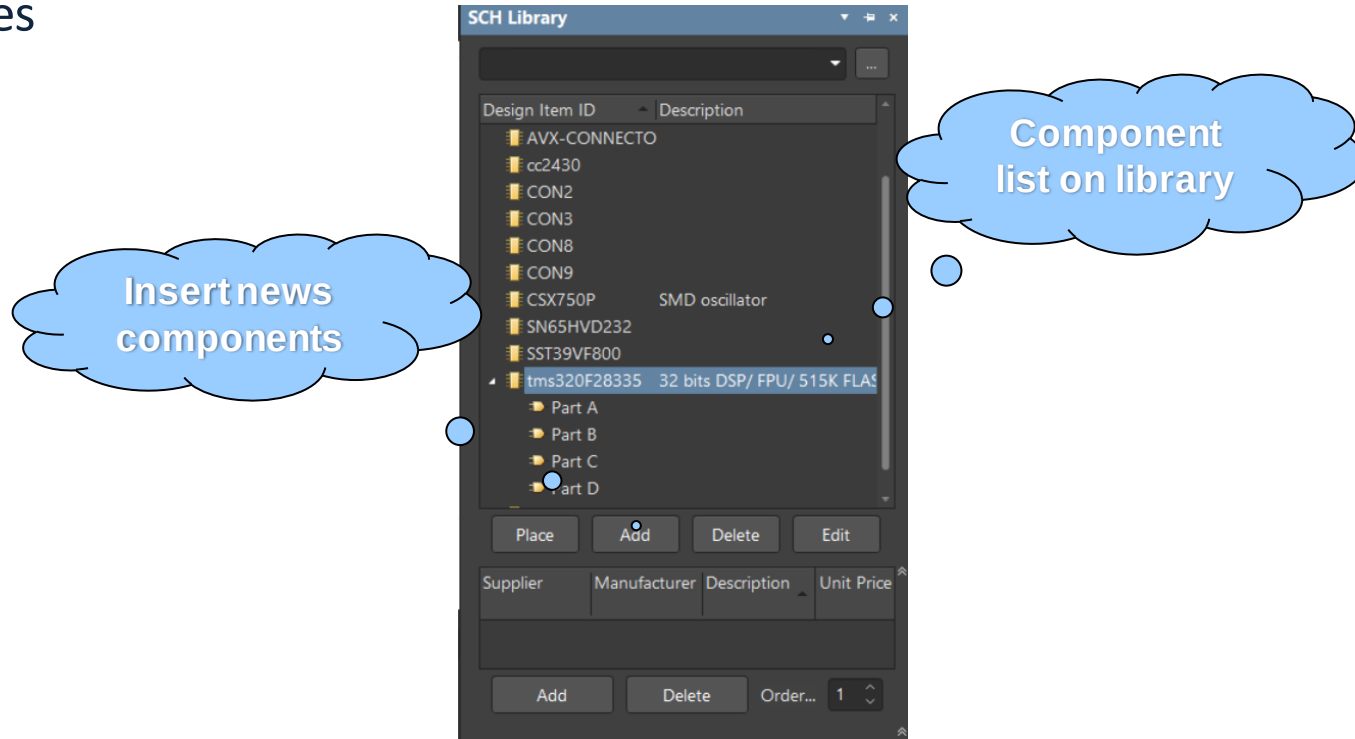
Use imperial units and set grid to 100 mils. This is the standard pin-to-pin distance in most libraries.



Do not forget
to set the grid

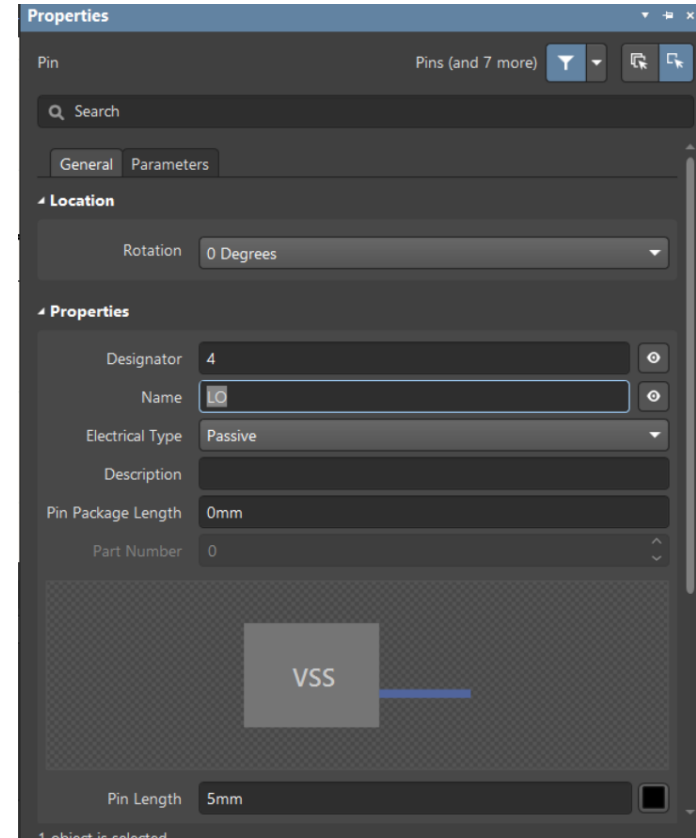
New Component

From the SCH Library panel you can always include new components or edit old ones



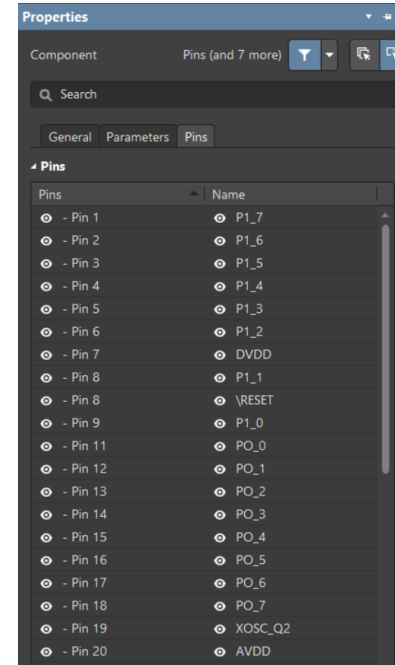
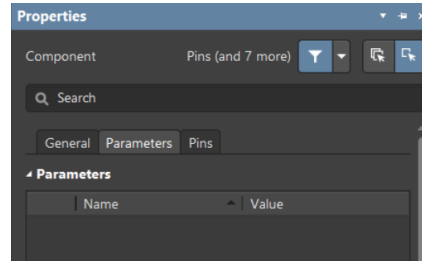
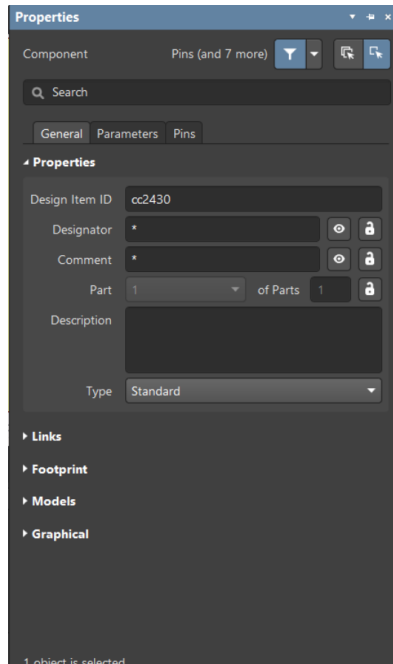
Adding pins

- **Place ➔ Pin** or select **Add** in the **Pins** section of the **SCH Library Panel**
- Before placing, press **TAB** to open properties panel
 - **Display name** is the name of the pin, and **Designator** its number
 - **Electrical type** for ERC
 - **Hide** to make hidden connection to power net
 - **Symbols** for clock, dot, etc.
 - **Length** and **Rotation** can be changed



Component Properties

From the Properties panel (you must deselect all the elements on component design), you can define all the information related with the component.

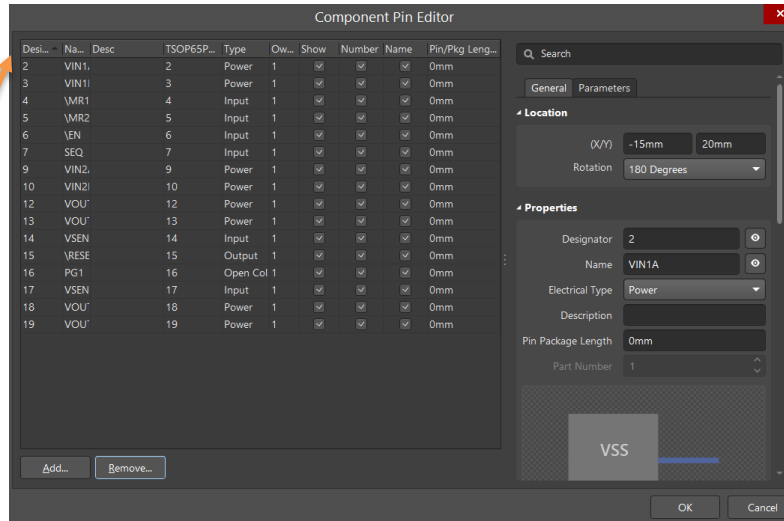
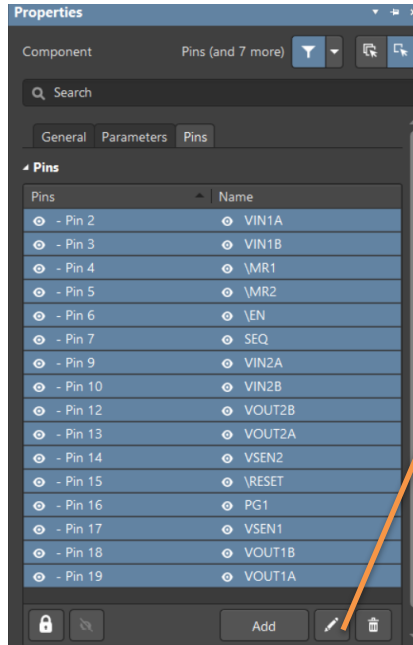


Component Properties

- Set the **Default Designator**: U?, R?, C?, etc.
- Set the **Default Comment** (part value)
- Write the optional **Description**
- Set the **Type** of the component:
 - **Standard**: These components possess standard electrical properties, are always synchronized and are the most common type on a schematic sheet.
 - **Mechanical**: These components do not have electrical properties and will appear in the BOM. They are synchronized if the same components exist on both the Schematic and PCB documents. An example is a heatsink.
 - **Graphical**: These components are not used during synchronization or checked for electrical errors. These components are used, for example, when adding company logos to documents.
 - **Tie Net in BOM**: These components short two or more different nets and these components will appear in the BOM and are maintained during synchronization.
 - **Tie Net**: These components short two or more different nets and these components will NOT appear in the BOM and are maintained during synchronization.
 - **Standard (No BOM)**: These components possess standard electrical properties and are synchronized BUT are not included in any BOM file produced from the file.
- Add **Parameters** and **Models**, at least add a footprint

Component Properties

The **Edit Pins** button of the **Component Properties** dialog is a more convenient alternative to add pins.



Spreadsheet based, press **Add** to add a new row (that is, a new pin) and specify its values by filling in the cells.

Importing a Pin List from Excel

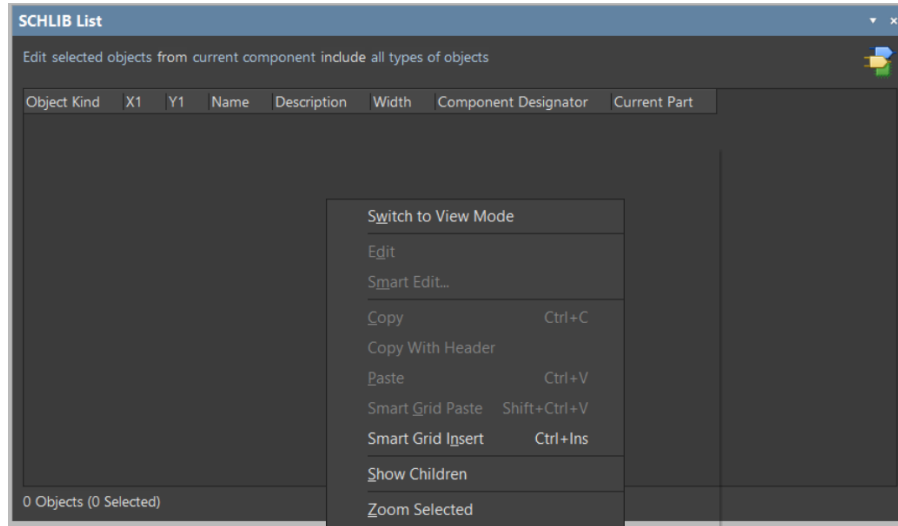
- Create a pin list in Excel with pin positions, orientations, names and designators.
- Add headers and a column “object kind” set to pin
- Example of the final aspect of the spreadsheet:

	A	B	C	D	E	F	G
1	Object Kind	X1	Y1	Orientation	Name	Pin Designator	
2	pin	100	100	180 degrees	pin1	1	
3	pin	100	200	180 degrees	pin2	2	
4	pin	100	300	180 degrees	pin3	3	
5	pin	100	400	180 degrees	pin4	4	
6	pin	800	100	0 degrees	pin5	5	
7	pin	800	200	0 degrees	pin6	6	
8	pin	800	300	0 degrees	pin7	7	
9	pin	800	400	0 degrees	pin8	8	
10							
11							

- When you are done, copy the whole list to the clipboard, including the header row

Importing a Pin List from Excel

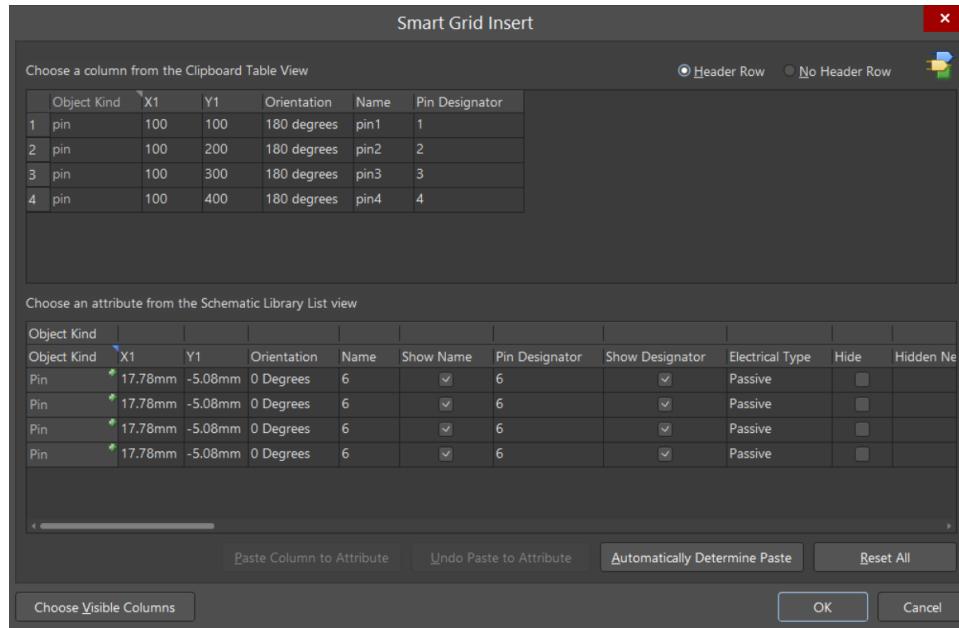
- Open the **SCH List** panel, and make sure that the mode is “Edit all objects from current component, include only pins”



- Right-click and select **Smart Grid Insert**

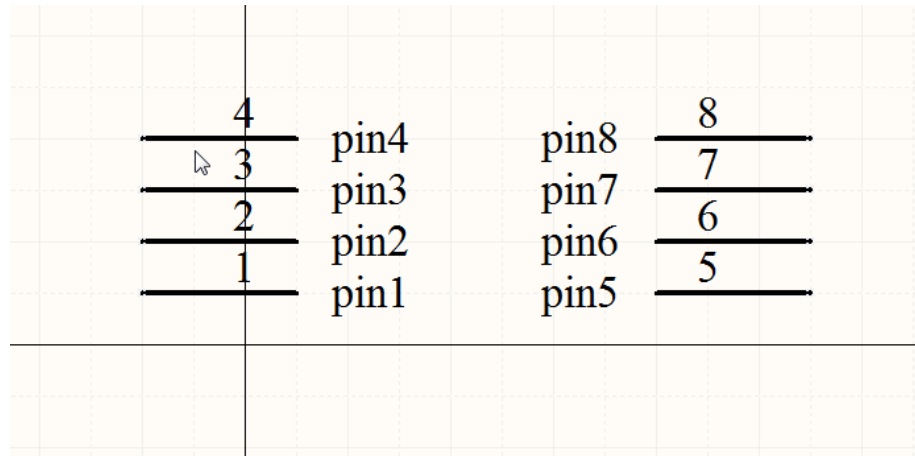
Importing a Pin List from Excel

Once the pin objects have been created, press **Automatically Determine Paste**, and make sure that changes are ok:



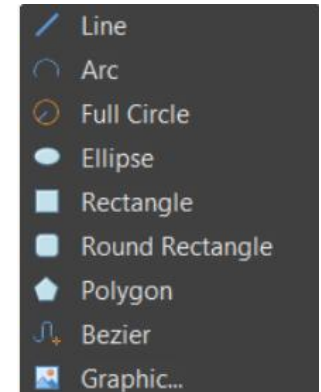
Importing a Pin List from Excel

Pinlist import is finished!



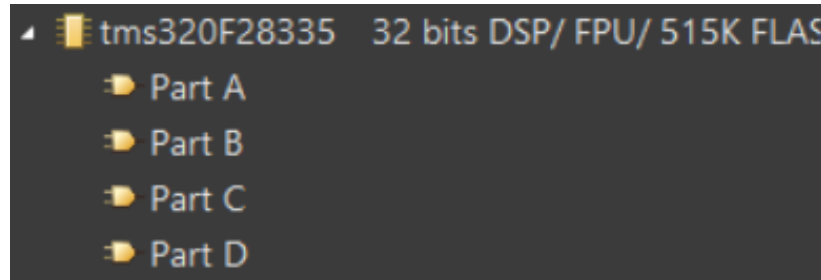
Drawing the outline

- After having created and positioned the pins, you must create the outline and any graphics you may want to add
- Use the design elements from the utilities toolbar
- The outline is simply a rectangle
- Any drawing objects can be added
- If the rectangle hides the pin numbers, remember that you can bring objects back and forward



Parts

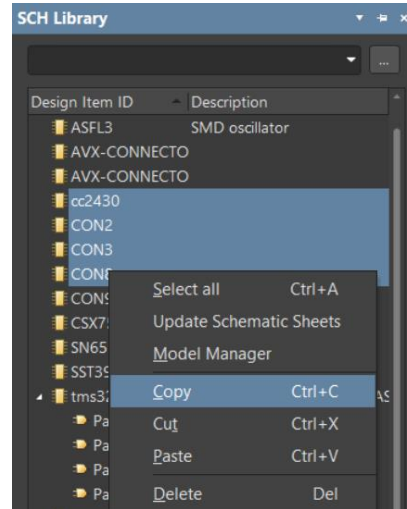
- You can create a **multipart package** by adding new parts to the component:
Tools ➔ New Part



- Typically, you will add a new part, copy the outline and drawing from the existing one, and add the new pins to the part
 - Remember that one of the properties of pin is the part to which it is associated

Copying from other libraries

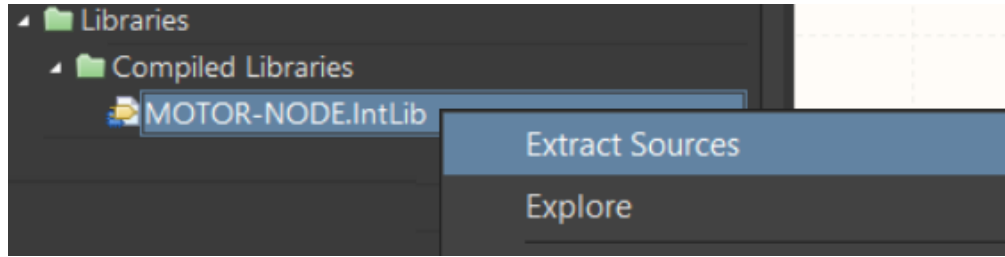
- Copying components from other library is as simple as opening the source library, right-clicking on the component in the SCH Library panel and select **Copy**



- Then open the target library and **Paste**

Breaking Integrated Libraries

- The problem is that **you cannot copy components from integrated libraries**
- **Simple solution:**
 - Open the integrated library (**File ➔ Open**)
 - Select **Extract Sources**, open the original schematic library. You can copy now
 - You can do that from a workspace, from right click

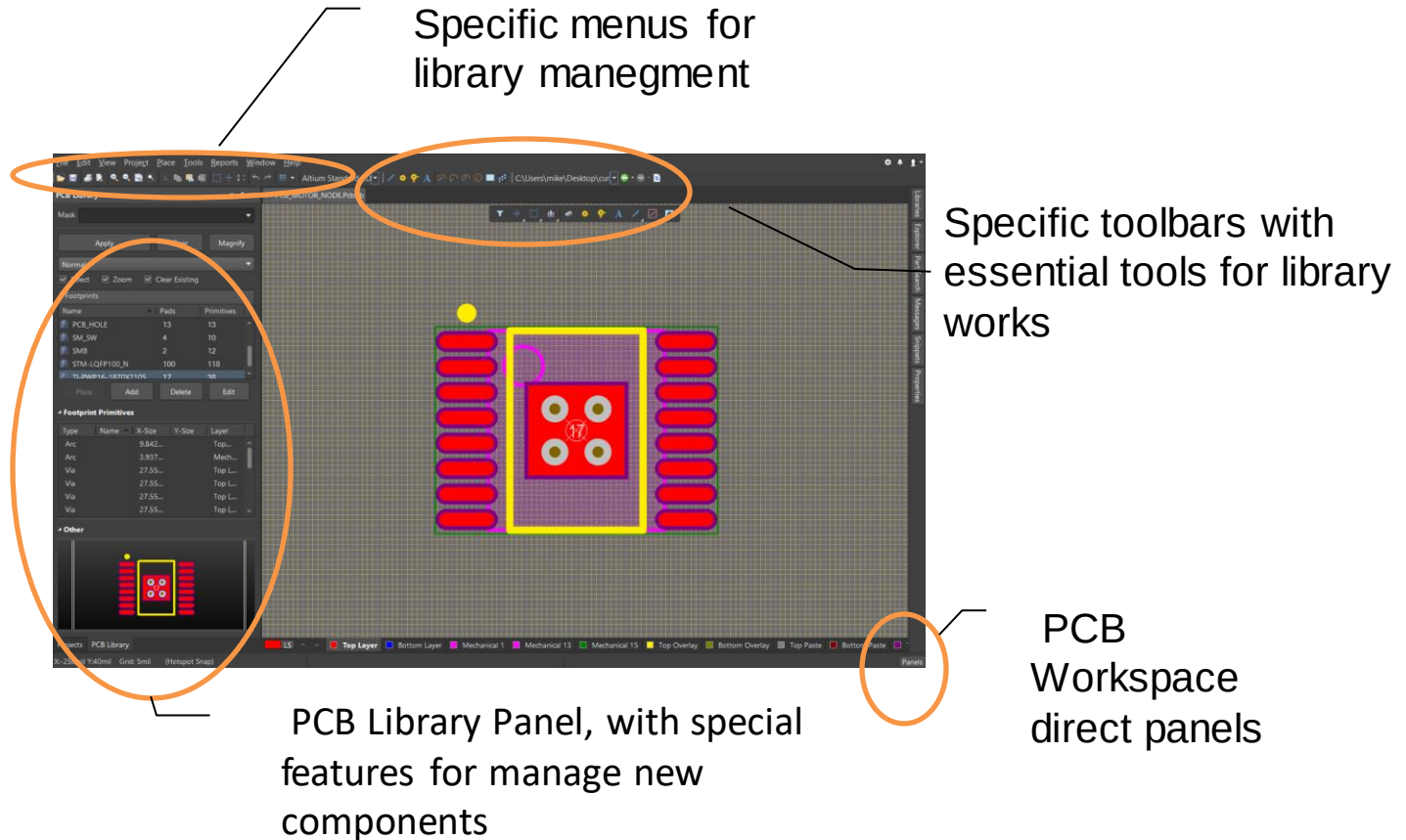




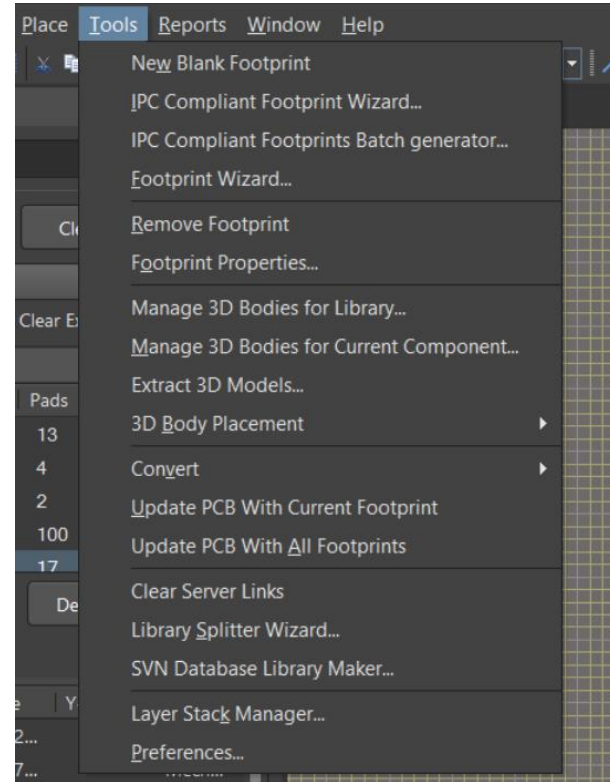
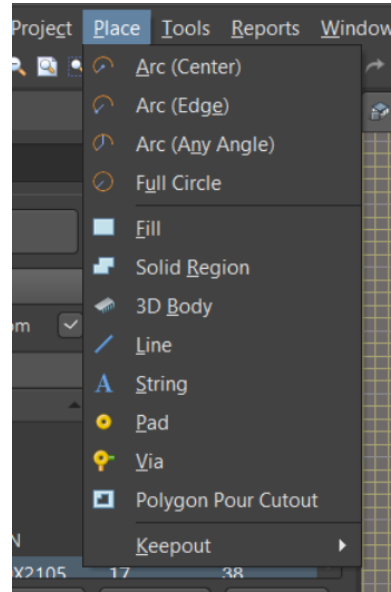
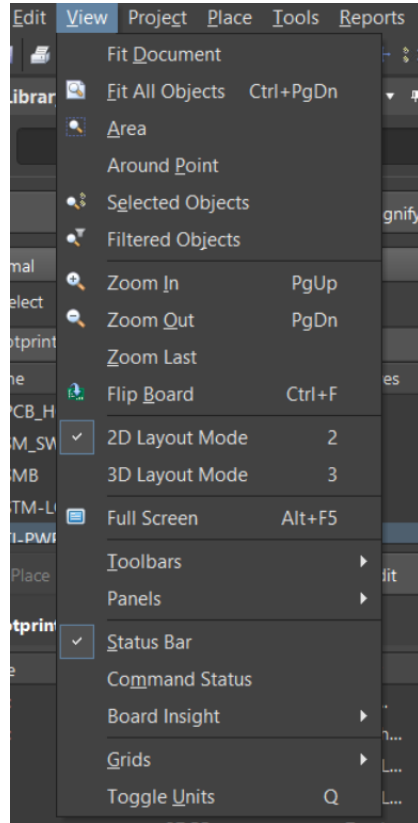
Agenda

- SCH libraries
- **PCB libraries**
- Via libraries

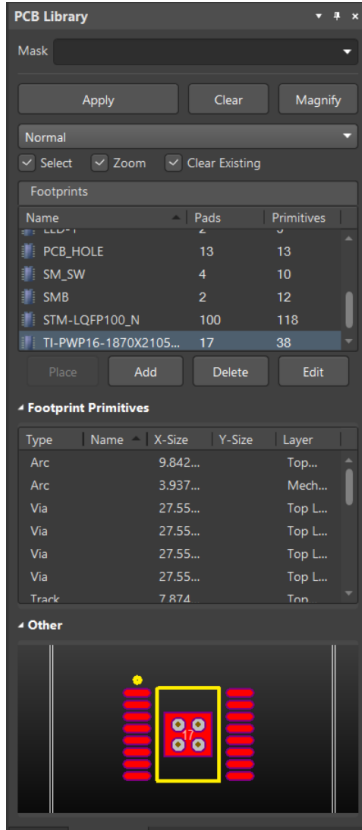
PCB Library



PCB Library



PCB Library

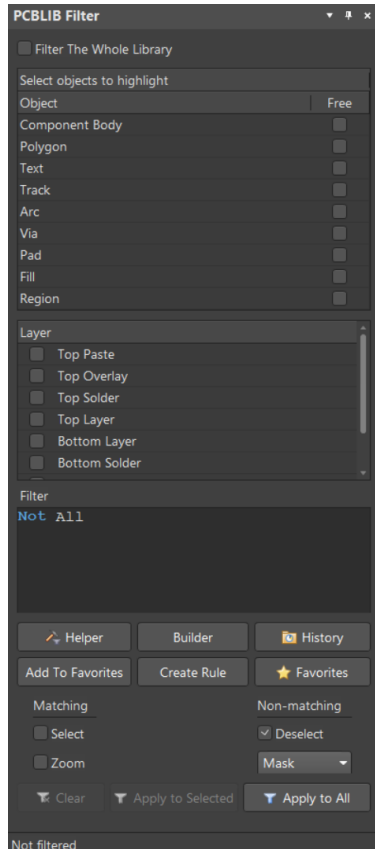


The PCB Library panel is divided into two main sections:

In the [Components section](#), we can find all the components in the active PCB Library. Right click in this section display a new menu where we can see creation, edition and update options.

The [Primitives sections](#) lists the primitives of the current selection component. Clicking on a primitive we can see it highlighted in the design window.

PCB Library

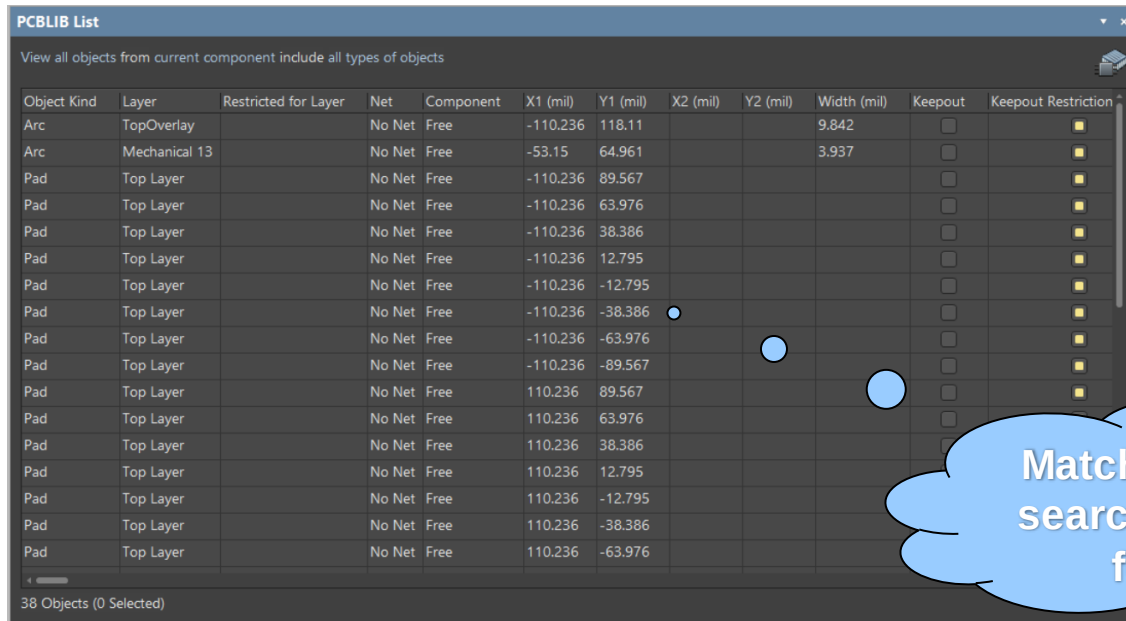


The **PCBLIB Filter** panel offers special to ease searching for special properties of a selected component.

We can use the Query language to apply filters.

PCB Library

One of the most unknown panels is the **PCBLIB List panel**, an efficient tool to work with the component.



The screenshot shows the 'PCBLIB List' window with a table of components. The table has columns for Object Kind, Layer, Restricted for Layer, Net, Component, X1 (mil), Y1 (mil), X2 (mil), Y2 (mil), Width (mil), Keepout, and Keepout Restriction. The data rows show various components like Arc and Pad with their respective coordinates and properties.

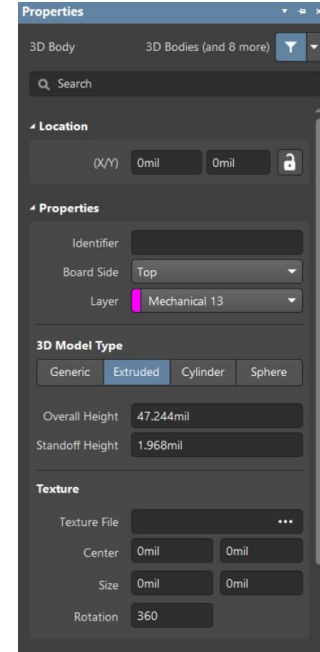
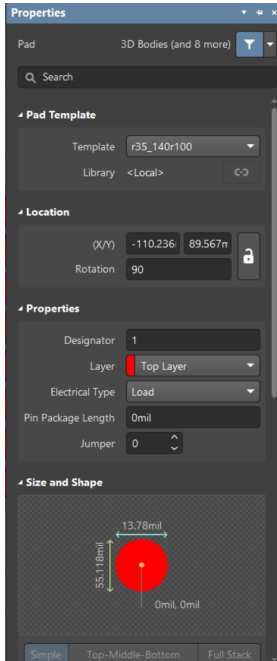
Object Kind	Layer	Restricted for Layer	Net	Component	X1 (mil)	Y1 (mil)	X2 (mil)	Y2 (mil)	Width (mil)	Keepout	Keepout Restriction
Arc	TopOverlay		No Net	Free	-110.236	118.11			9.842	☐	☐
Arc	Mechanical 13		No Net	Free	-53.15	64.961			3.937	☐	☐
Pad	Top Layer		No Net	Free	-110.236	89.567				☐	☐
Pad	Top Layer		No Net	Free	-110.236	63.976				☐	☐
Pad	Top Layer		No Net	Free	-110.236	38.386				☐	☐
Pad	Top Layer		No Net	Free	-110.236	12.795				☐	☐
Pad	Top Layer		No Net	Free	-110.236	-12.795				☐	☐
Pad	Top Layer		No Net	Free	-110.236	-38.386				☐	☐
Pad	Top Layer		No Net	Free	-110.236	-63.976				☐	☐
Pad	Top Layer		No Net	Free	-110.236	-89.567				☐	☐
Pad	Top Layer		No Net	Free	110.236	89.567				☐	☐
Pad	Top Layer		No Net	Free	110.236	63.976				☐	☐
Pad	Top Layer		No Net	Free	110.236	38.386				☐	☐
Pad	Top Layer		No Net	Free	110.236	12.795				☐	☐
Pad	Top Layer		No Net	Free	110.236	-12.795				☐	☐
Pad	Top Layer		No Net	Free	110.236	-38.386				☐	☐
Pad	Top Layer		No Net	Free	110.236	-63.976				☐	☐

38 Objects (0 Selected)

Matching tool for
searching special
features

PCB Library

The **Properties panel** shows all the information about the selected object.



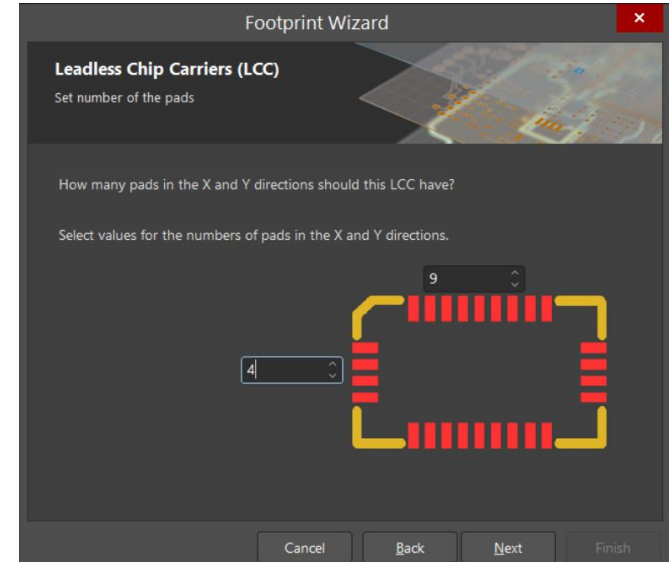
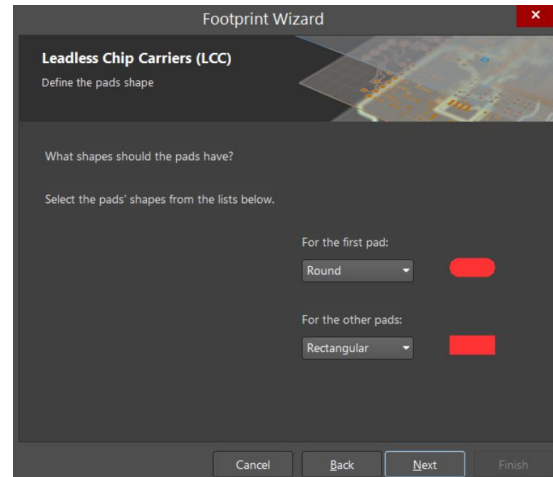
PCB Library

Tools

- New Blank Footprint
- IPC Compliant Footprint Wizard...
- IPC Compliant Footprints Batch generator...
- Footprint Wizard...**
- Remove Footprint
- Footprint Properties...

Component Wizard gives the designer an easy way to define a footprint, with basic information about the component.

We can choose from a wide range of footprint types (DIP, BGA, QFN...) with special submenus.



PCB Library

IPC Compliant Footprint Wizard offers specially defined submenus based on the international IPC-7351 rules.

Under the constraints of the official IC datasheets, we can make sure that the resulting footprint is IPC compliant.

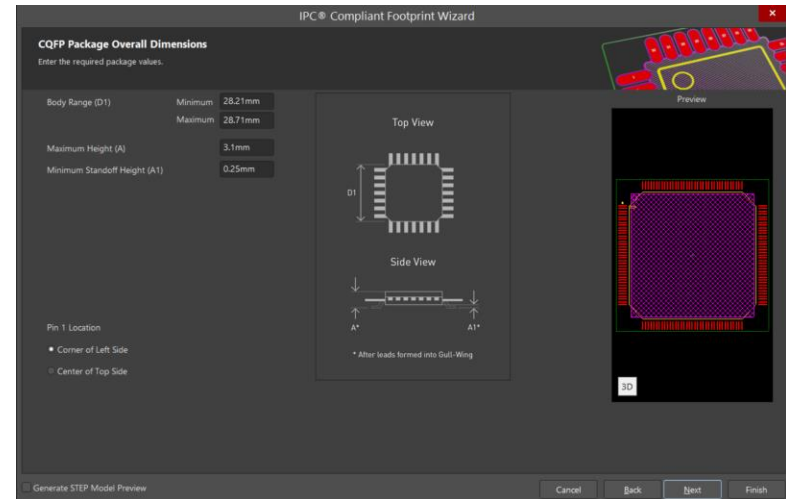
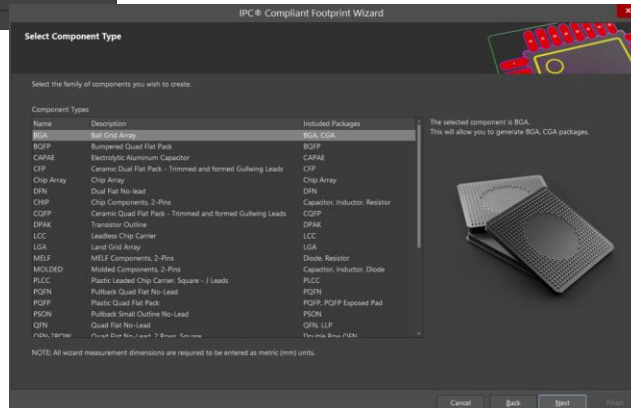
Tools

New Blank Footprint

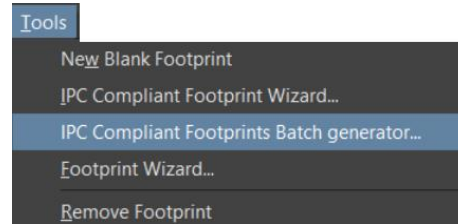
IPC Compliant Footprint Wizard...

IPC Compliant Footprints Batch generator...

Footprint Wizard...



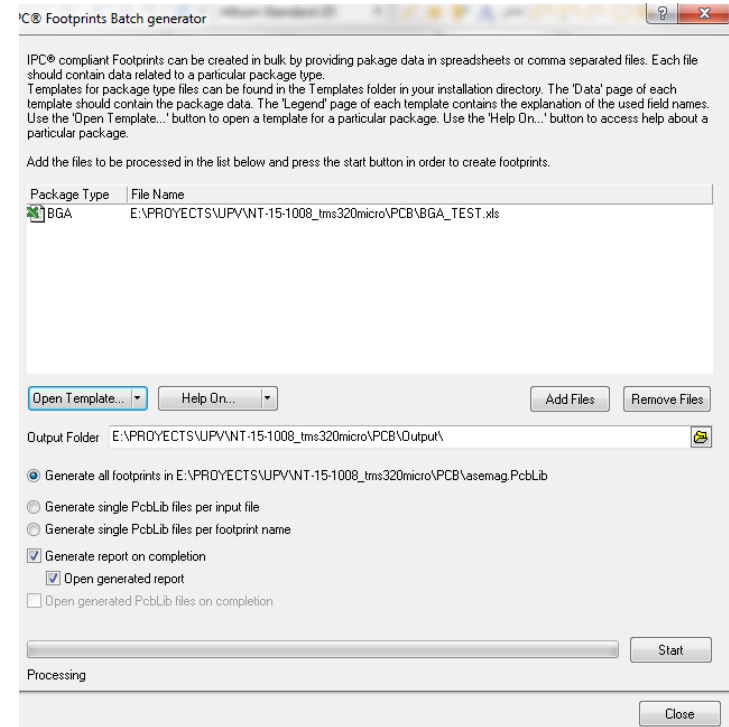
PCB Library



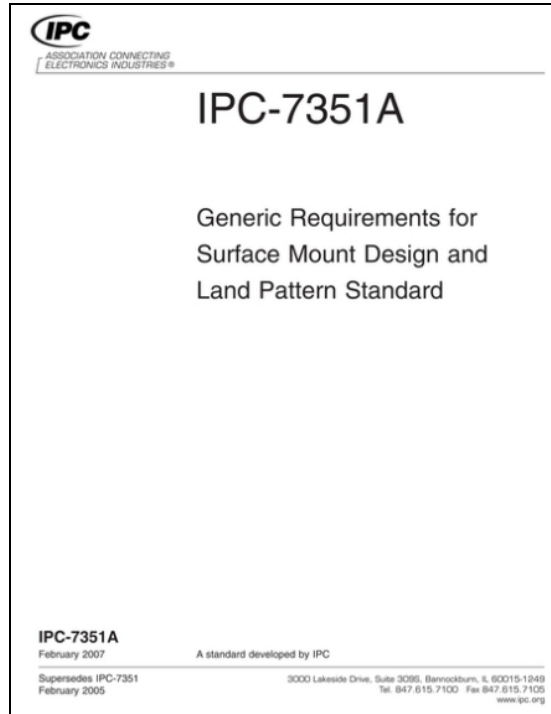
In the same way of IPC Wizard, with **IPC Batch generator** we obtain an IPC footprint compliant with IPC rules.

This batch generator offers a tool focused in providing package data as a description, usually from an Excel file.

IPC Batch generator increases efficiency creating a large number of footprints from a single file.



PCB Library



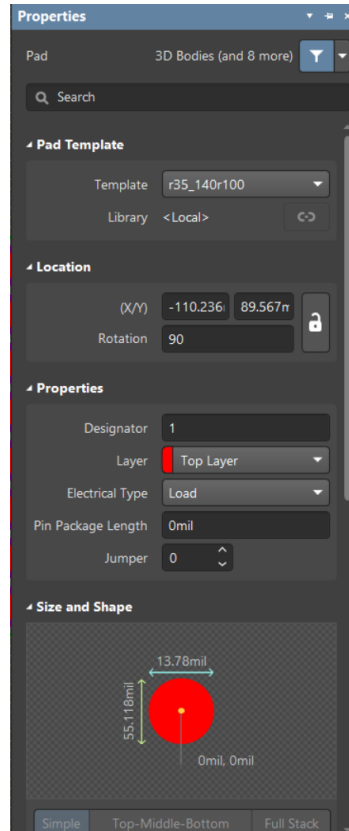
www.ipc.org

The **IPC-7351A rules** establish the generic requirements for the SMD design and other PCB component footprints (active and passive comp., MELF, SSOP, TSSOP, QFP, BGA, QFN and SON).

The IPC rules use dimensional information to generate a valid footprint. This footprint is based on dimension tolerances from the component manufacturer.

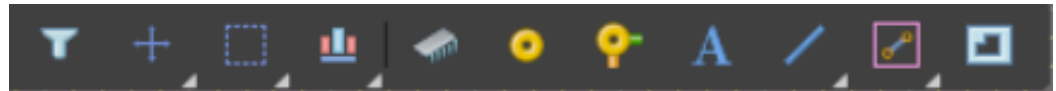
These rules avoid the most usual mistakes associated with component mounting.

PCB Library



The basic tools for manual working are **lines**, **arcs** and the use of the **Pad tool**.

With the pad tool we can define or edit any pad in the design. We can define multilayer pads, holes, dimensions, solder mask, etc.

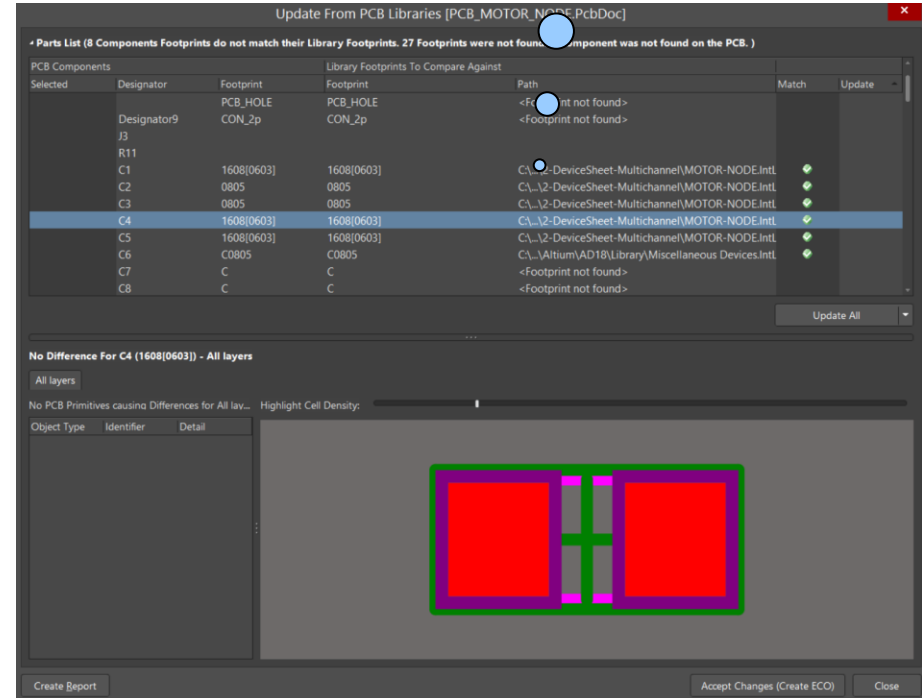
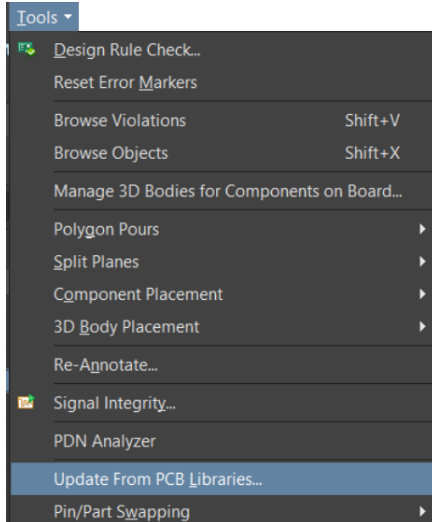


Press Tab from the keypad to modify object properties.

PCB Library

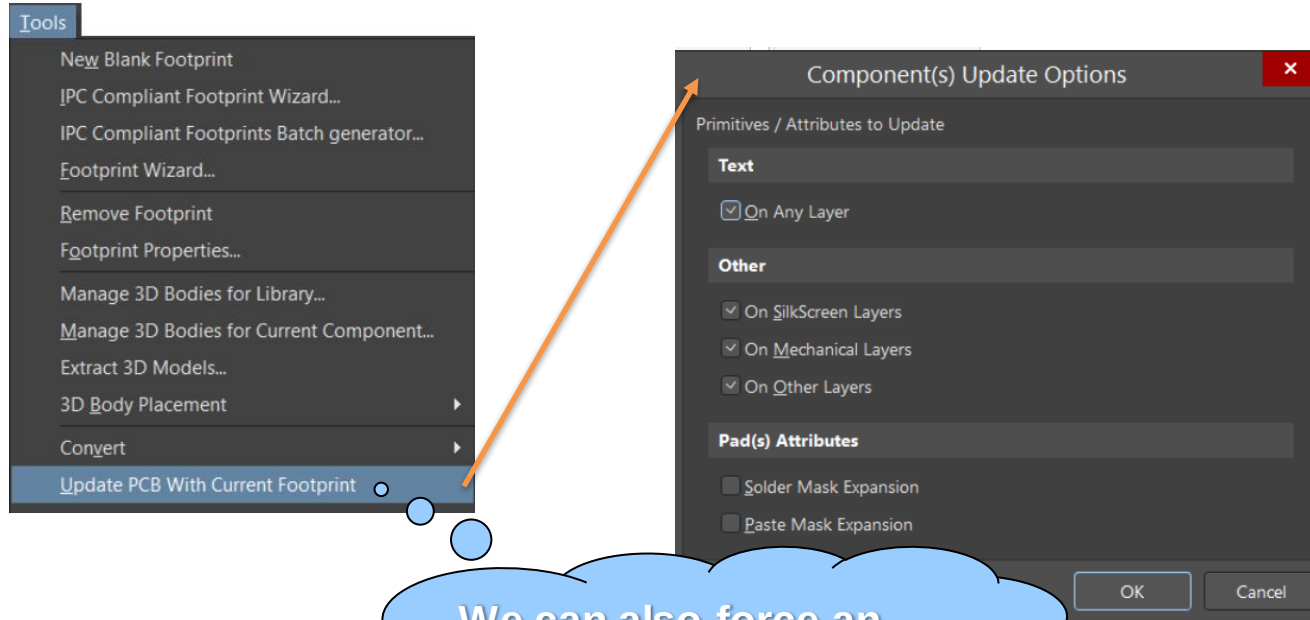
Update from PCB Libraries

Matching their
differences with
libraries



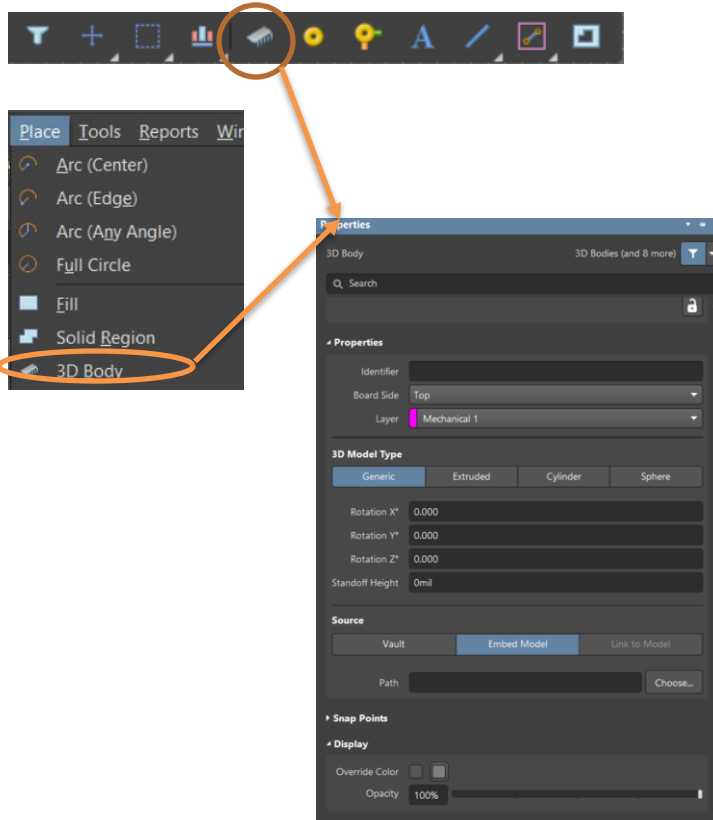
PCB Library

Update PCB with current footprint



We can also force an update from the library

PCB Library



3D Body tool

Manual definitions are made from Extrude, Cylinder and Sphere options.

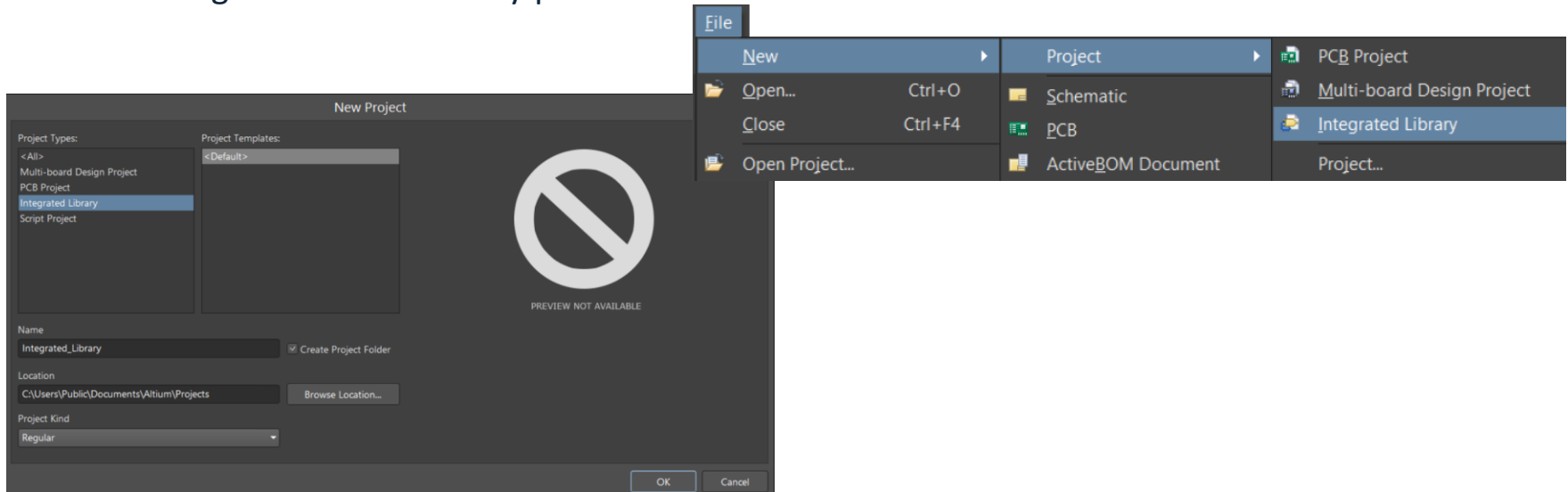
With **Generic Model** option we can import external model based on **STEP** model file.

If we choose the **extruded option**, we can do the 3D body from scratch.

Integrated libraries

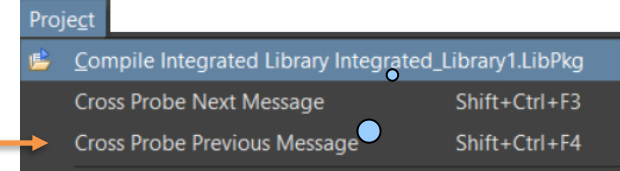
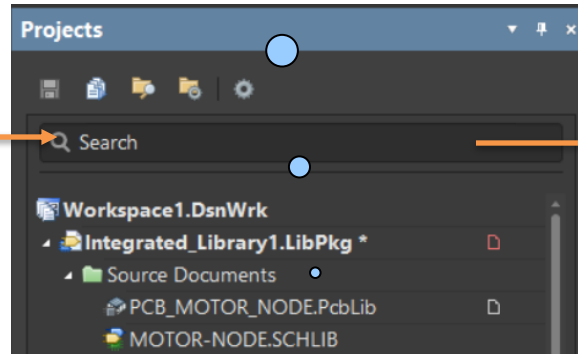
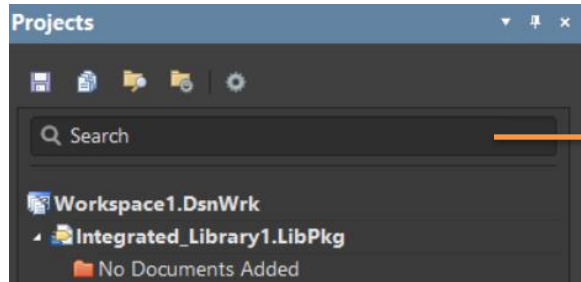
Combine schematic libraries with their related PCB footprints and/or SPICE and signal integrity models, all together in a non-editable form.

Model information is copied into the integrated library from the model libraries or files and so all the component information is stored together, regardless of the location of the original source libraries. This makes integrated libraries truly portable.



Integrated libraries

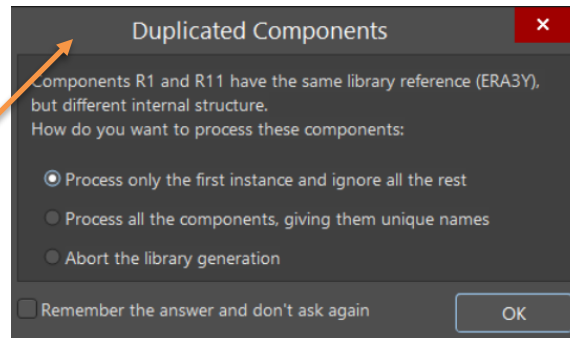
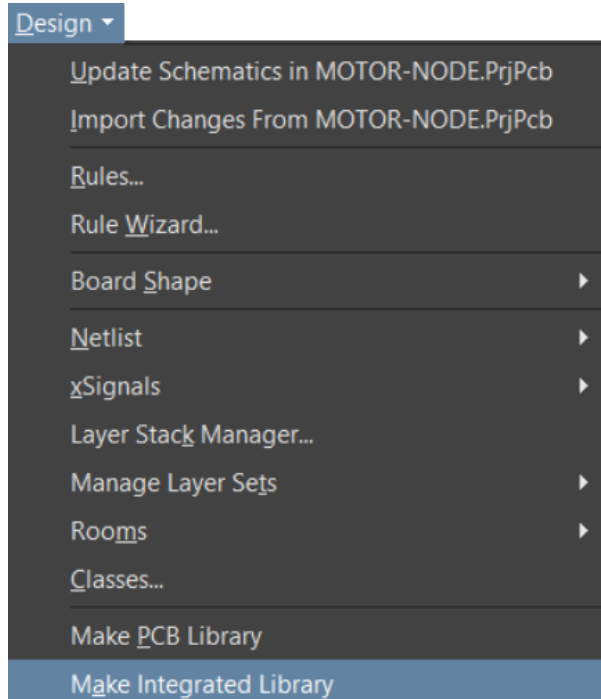
Once we have an Integrated Lib. Project we can include their libraries



To finish the Integrated Lib. only need to compile and save

Integrated libraries

You can create a schematic library out of the components that have been already placed on schematic or PCB documents in a project using the Make Integrated\SCH\PCB Library commands





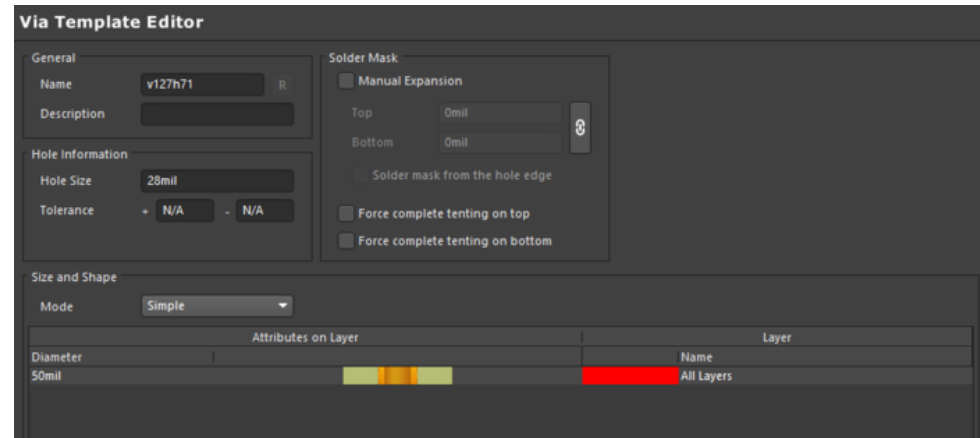
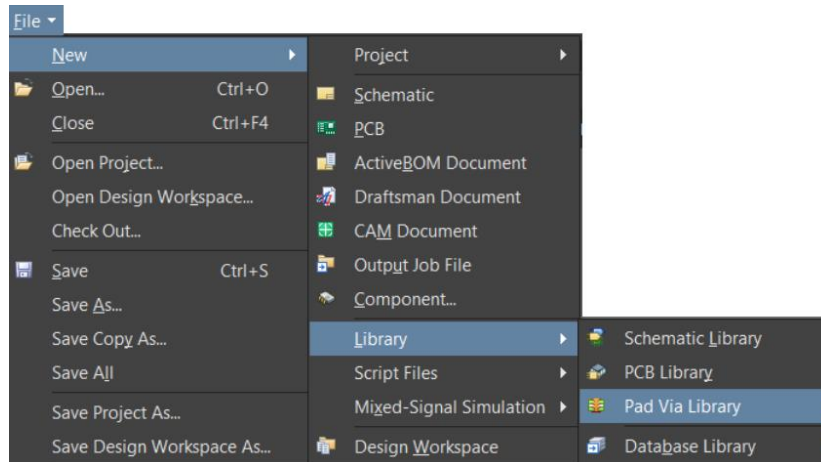
Agenda

- SCH libraries
- PCB libraries
- **Via libraries**

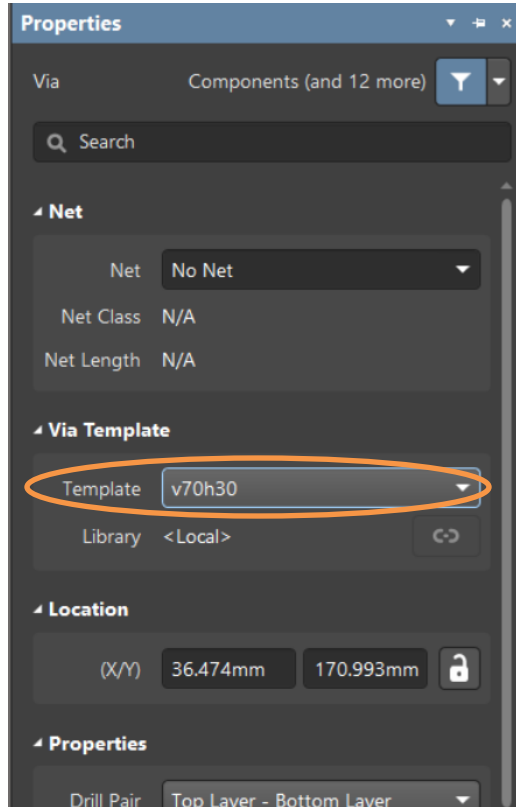
Via libraries

The concept of **Pad and Via templates** that can be collected into a Library is like that of PCB footprint Libraries, although somewhat more basic. The templates are composed of primitive design elements with defined properties, and these can populate an associated library.

This saved library can be reloaded and used to place instances of predefined Pads and Vias in any PCB design (or PCB Library).



Via libraries

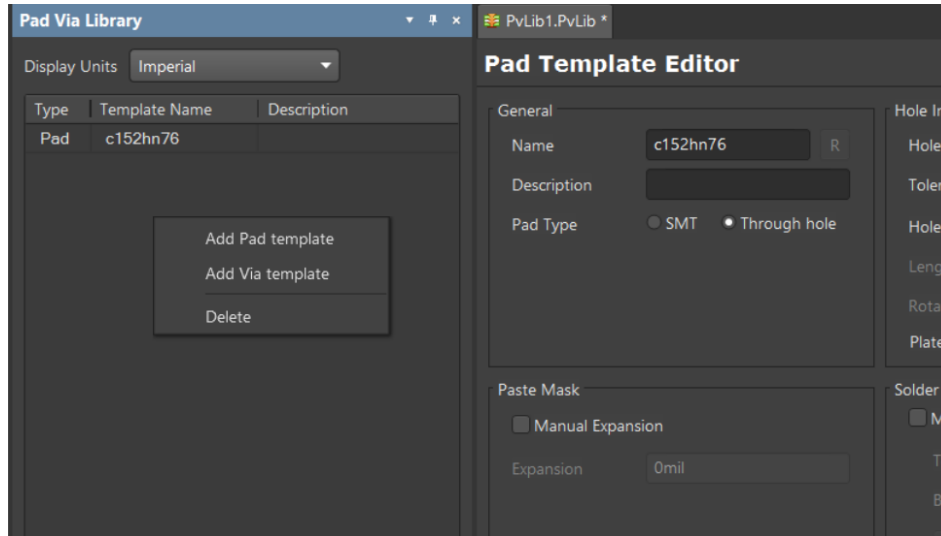


Pad and Via types that are present in an existing PCB design are automatically named when the design file is opened, ensuring compliance with the new functionality.

To observe this behavior, inspect the properties of an existing Pad or Via from its associated *Properties* dialog – double click on the object, or select the object then choose **Properties** from Tab keypad.

Via libraries

To include a new pad or via, you only have to right click and select what to include in your pad-via library.



The IPC naming system is metric based where one unit is equivalent to 10 μm .

For example, the template for a 1.5 mm circular pad with a 0.8 mm hole is named c150h80 – where c indicates circular (round) and h prefixes the hole size.



Agenda

FIN DEL MÓDULO